JERIN JACK JOSEPH RAJ

Software Engineer | Java, Spring Boot & Microservices

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years building scalable Spring Boot microservices and RESTful APIs. Strong in Java 8/11/17, SOLID design, and relational and NoSQL database design across Oracle, MySQL, PostgreSQL, and MongoDB. Experienced with Kafka and JMS event-driven messaging, Spring Security with OAuth2 and JWT, and CI/CD with Jenkins, GitHub Actions, Docker, and AWS ECS. Works AI-natively: AI coding assistants are part of the daily workflow for code generation, test scaffolding, and documentation. Track record of debugging production issues and making slow systems fast.

TECHNICAL SKILLS

Languages & Core: Java 8/11/17, Core Java, OOP, SOLID Principles, SQL
APIs & Microservices: RESTful APIs, Microservices Architecture, API Design, gRPC (working knowledge)
Frameworks & Security: Spring Boot, Spring MVC, Spring Security, Spring Data, Hibernate/JPA, OAuth2, JWT
Databases: Oracle (PL/SQL, query tuning), MySQL, PostgreSQL, MongoDB, indexing, SQL optimization
Messaging: Apache Kafka, JMS, event-driven architecture
Cloud & DevOps: AWS SDK, Azure SDK, Docker, Kubernetes (working knowledge), Jenkins, GitHub Actions, CI/CD Pipelines
Testing & Quality: JUnit, Mockito, Postman, SonarQube, code reviews, root cause analysis
AI-Native Development: AI coding assistants for code generation, test scaffolding, and documentation
Frontend: Angular 8, HTML5, CSS3, TypeScript
Methods: Agile/Scrum, Jira, sprint planning, technical writing

PROFESSIONAL EXPERIENCE

AVPS Technologies Inc Software Engineer | Jul 2026 – Present
Alpharetta, GA

- Delivered 12+ production REST endpoints for the company's content and analytics platform by designing and developing Java 8/17 Spring Boot microservices with a clean controller-service-repository architecture and reusable DTOs.
- Reduced response times on high-traffic endpoints by approximately 25% by resolving a Hibernate mapping issue, adding missing database indexes, and optimizing SQL queries against Oracle and PostgreSQL.
- Reduced service coupling and improved reliability by implementing Kafka and JMS-based asynchronous processing for audit events and workflow notifications.
- Maintained zero production defects over the past year by conducting code reviews and mentoring two junior developers on engineering standards.
- Reduced manual deployment effort by managing CI/CD pipelines in Jenkins and GitHub Actions, containerizing services with Docker, and deploying to AWS ECS using AWS and Azure SDKs.
- Improved day-to-day development velocity by applying AI coding assistants to boilerplate code generation, test scaffolding, and documentation.
- Supported end-to-end application workflows by integrating backend services with the Angular front end and downstream systems.

Chicago Education Advocacy Cooperative Software Engineer | Aug 2025 – Jul 2026
Chicago, IL

- Built out the backend for the organization's content and internal operations tools by developing a set of Java 17 Spring Boot services with reusable DTO contracts.
- Cut a recurring reporting query from several seconds to under a second by adding the right database indexes and rewriting an inefficient join.
- Reduced dependency on synchronous service calls by implementing Kafka-based asynchronous processing for audit events and workflow updates.
- Improved release quality by writing unit and integration tests using JUnit, Mockito, and Postman, and raising SonarQube code coverage ahead of releases.
- Resolved Tier 3 production issues faster by using Log4j output and API traces for diagnosis and root cause analysis.
- Maintained consistent, reliable deployments by managing CI/CD pipelines with GitHub Actions,

Jenkins, and Docker on AWS ECS and Kubernetes-compatible infrastructure.
- Supported cross-team API integration by producing OpenAPI-style documentation alongside PostgreSQL schema and query development.
- Contributed to consistent sprint delivery by participating in sprint planning, story estimation, and code review with QA, DevOps, and product teams.

Gulf International Bank (via TCS) Backend Developer | Oct 2021 – Jul 2023
Chennai, India
- Delivered core backend services for balance transfers, international payments, account lookups, and reconciliation by developing Java and Spring Boot applications for a regulated banking client.
- Cut peak-time payment API response times nearly in half by tuning Oracle/PL-SQL queries, adding indexes, managing transactions, and implementing Redis caching.
- Enabled secure REST and SOAP service integration by building request validation, response mapping, and error-code handling with Spring MVC and Hibernate/JPA.
- Extended payment processing reliability by integrating internal systems with partner APIs and core banking platforms through WSO2 API Manager, JMS, and Kafka event flows.
- Supported nightly reconciliation and business reporting by building Apache Spark batch jobs and ETL pipelines feeding Tableau dashboards.
- Reduced release risk by conducting code reviews, mentoring junior developers, and maintaining Jenkins CI/CD pipelines, with sprint tracking in Jira.

Iota Automation Private Limited Java Developer | Apr 2020 – Jul 2021
Chennai, India
- Delivered REST APIs and backend services for internal enterprise applications by developing Java and Spring Boot solutions on Oracle and MySQL.
- Improved code maintainability by applying SOLID principles and standard design patterns, including DTOs, the repository pattern, and centralized exception handling.
- Sped up high-read reporting operations by optimizing queries and adding targeted database indexes.
- Reduced regression risk by writing unit and service-level tests using JUnit and Mockito and validating APIs with Postman.
- Supported production deployments by managing Linux/Unix environments, Tomcat configuration, and shell scripts for build validation.

RELEVANT PROJECTS

Vehicle Reservation Platform: Spring Boot Microservices & Angular
- Built an end-to-end vehicle reservation platform with Java/Spring Boot microservices and an Angular 8 front end for search, booking, and an administrative dashboard.
- Enabled real-time booking updates by integrating GIS-based location mapping and Azure IoT Hub/Event Hub for vehicle telemetry.
- Secured the platform's API layer with JWT authentication and role-based access control, with JUnit/Mockito test coverage, deployed via Docker and Kubernetes through Jenkins CI/CD.
- Built AI-natively: AI coding assistants for boilerplate, test scaffolding, and documentation.

AI-Powered Document Processing Platform: Spring Boot & AI
- Automated processing of invoices, IDs, and contracts with a Spring Boot service using AI-based text extraction and NLP classification models.
- Improved processing throughput with Kafka-based asynchronous job queuing, with results stored across MongoDB and Oracle by document type.
- Exposed REST APIs for upload, status tracking, and result retrieval for downstream consumers, backed by unit/integration tests and SonarQube quality gates.

Real-Time Analytics & ETL Reporting Pipeline: Spring Boot, Kafka & Spark
- Enabled real-time business reporting with Spring Boot services driving a Kafka and Spark streaming pipeline into Oracle-based reporting tables.
- Delivered stakeholder-facing operational dashboards by building ETL jobs to clean, transform, and load multi-source data, published via Tableau.
- Built AI-natively: AI coding assistants for boilerplate, test scaffolding, and documentation.

EDUCATION

M.S. Information Technology and Management, Illinois Institute of Technology, Chicago, IL | GPA: 3.9/4.0 | Aug 2023 – May 2025

B.E. Electrical and Electronics Engineering, Anna University, Chennai, India | GPA: 3.4 | Aug 2017 – 2021